

Unit-5

- **XML & WEB DEPLOYMENT**
- Concept of DOMAIN , Concept of Hosting ,Deployment using FTP ,PHP FRAMEWORK USING WORDPRESS

- **XML (Extensible Markup Language)**
 - **What is XML?**
 - **XML** stands for **Extensible Markup Language**.
 - It is used to **store and transport data** in a structured way.
 - It looks similar to HTML, but **XML focuses on data**, while **HTML focuses on display**.
-
- **Why XML?**
 - It helps in **sharing data** between different applications or systems.
 - Data is **human-readable** and **machine-readable**.
 - Commonly used in web services, APIs, and configuration files.

- **Example of XML**

- <student>
- <name>John</name>
- <age>21</age>
- <department>Computer Science</department>
- </student>

Explanation:

- <student> is the root tag.
- Each data field is stored inside tags (<name>, <age>, <department>).
- XML does not define how data looks — it only defines **what the data is**.

- **Where XML is Used:**

- Configuration files (e.g., web.xml in Java web apps)
- Data exchange between systems
- Web services (SOAP uses XML)
- Android apps (AndroidManifest.xml, layout XML files)

- **Web Deployment**
- **What is Deployment?**
- **Deployment** means **making your website or application available on the internet** so that users can access it through a web browser.

Steps in Web Deployment

- **Develop the Website**
 - Create HTML, CSS, JS, or PHP files on your local computer.
- **Test Locally**
 - Use tools like **XAMPP/WAMP** to check your PHP/MySQL pages locally.
- **Get a Domain and Hosting**
 - **Domain Name:** Your website address (e.g., www.mycollegeproject.com)
 - **Web Hosting:** A server space where you upload your files.
- **Upload Files to Server**
 - Use **FTP (File Transfer Protocol)** to transfer your project files to the hosting server.
 - Tools: **FileZilla, cPanel File Manager.**
- **Database Setup**
 - Import your MySQL database to the hosting server using **phpMyAdmin.**
- **Test Online**
 - Visit your domain to check if the website runs correctly.

Example of Deployment Tools:

Tool

FileZilla

cPanel

XAMPP/WAMP

Purpose

Upload files to hosting server

Manage hosting, databases, and files

Local testing before deployment

- XML is primarily used to **store and transport data** in a structured, self-describing, and universal format.
- It is not a programming language, so you don't "run" an XML file; instead, it is **processed** by other applications

- XML (eXtensible Markup Language) is essential for data exchange between different systems that might use incompatible formats.
- Data Transport & Exchange: This is its main purpose. It provides a common, structured format that can be easily read by any application (e.g., a website, a mobile app, or a database system) regardless of the underlying programming language or operating system.
- Self-Describing Data: XML uses user-defined tags (like <to>, <from>, <message>) to describe what the data is. This makes the data human-readable and allows a program to understand the context of the data without needing an external dictionary.

- Configuration Files: XML is often used to store settings and configurations for software applications (e.g., in Android development or older Microsoft Office file formats like .docx).
- Web Services: Protocols like SOAP (Simple Object Access Protocol) rely on XML to structure the messages sent between clients and servers.

How to "Run" (View and Process) an XML File

- Since XML is just structured text, you view it in an editor and use a separate program to process its data.

Method	Description	Advantage
Text Editor	Right-click the file and choose Open With > Notepad (Windows) or TextEdit (Mac).	Shows the pure, underlying code for editing.
Web Browser	Drag and drop the .xml file into Chrome, Firefox, or Edge.	The browser often displays the file with a collapsible, tree-like structure , making the hierarchy easier to read.
Code Editor	Use specialized editors like VS Code or Sublime Text .	Provides syntax highlighting and error checking. ¹⁰

- The connection between **XML (eXtensible Markup Language)** and **Web Deployment** is primarily centered on how XML is used to **configure, describe, and transfer data** essential for setting up and running web applications.

- **Configuration and Deployment Descriptors**
- The most significant use of XML in web deployment is for **configuration files** that instruct the server or application runtime environment on how to behave.
- **Java Web Applications (e.g., Tomcat, JBoss):**
 - **web.xml (Deployment Descriptor):** This is a core XML file used in Java EE (now Jakarta EE) applications. It defines crucial deployment details like:
 - **Servlets and their mappings** (which URL maps to which Java class).
 - **Security constraints** (who can access certain pages).
 - **Context parameters** (global settings for the application).
 - **Welcome files and error pages.**
 - **Server Configuration:** Files like **server.xml** in Tomcat use XML to configure connectors, virtual hosts, and data sources.
- **Microsoft .NET (Older Deployments):**
- **web.config:** This XML file is used in ASP.NET applications to configure settings like connection strings, security settings, and logging providers.

- **Data Exchange and Packaging**

- XML helps package and transport the deployment information itself.
- **Metadata:** XML is used to create **manifests** or **metadata files** that describe the contents of a deployment package (e.g., a .war or .jar file). This helps the deployment tool verify and unpack the application correctly.
- **Web Services (SOAP):** If your web application provides or consumes **SOAP (Simple Object Access Protocol)** services, all communication—the requests and responses—are formatted as XML. This XML data is transported over HTTP, enabling different systems to interact.

- **Automated and Scripted Deployment**
- XML provides a structured, machine-readable format for automation tools.
- **Ant and Maven:** Build and deployment tools like **Apache Ant** and **Apache Maven** use XML files (build.xml and pom.xml, respectively) to define the **build process**, dependencies, and the **deployment targets**. This allows deployment to be automated and repeatable across different environments.
- **Templating:** Deployment processes often use XML as a template. Scripts can easily read the XML structure, replace placeholder values (like database passwords or server hostnames), and output the final configuration file for the target environment.
- In summary, while XML doesn't execute the deployment itself, it serves as the **universal language** for storing the necessary **instructions, configurations, and data descriptions** that deployment systems need to set up and run the web application successfully.

- **What is a Domain?**
- A **domain** is the **name of a website** that users type in a browser to visit a site.
It represents the website's **address on the Internet**.
- **Example:**
- Domain: `www.google.com`
- Here, `google.com` is the domain name.
- **Key Points:**
- Domains are easier to remember than IP addresses.
- A domain has two main parts:
 - **Second-Level Domain (SLD):** The unique name (e.g., `google`)
 - **Top-Level Domain (TLD):** The extension (e.g., `.com`, `.org`, `.edu`)

Example Structure:

- https://www.example.com
- ↑ ↑
- Domain TLD

Common Domain Types:

Type	Example	Usage
.com	google.com	Commercial websites
.org	wikipedia.org	Organizations
.edu	mit.edu	Educational institutions
.in	nic.in	Indian websites
.gov	india.gov.in	Government websites

Concept of Hosting

- **What is Web Hosting?**
- **Web Hosting** is a **service that stores your website files** (HTML, PHP, images, etc.) on a **server** so they can be accessed online.
- When a user visits your domain, the hosting server **sends the website content** to their browser.

Types of Hosting:

Type	Description	Example Use
Shared Hosting	Many websites share one server	Small websites
VPS (Virtual Private Server)	Shared but isolated sections	Medium traffic sites
Dedicated Hosting	One website per server	Large business sites
Cloud Hosting	Uses multiple servers	Scalable applications

- **Hosting Providers:**

- Bluehost
- Hostinger
- GoDaddy
- AWS (Amazon Web Services)

3. Deployment using FTP

- **What is FTP?**
- **FTP (File Transfer Protocol)** is a method used to **upload website files** from your computer to a **web server**.
- **Tools Used:**
- **FileZilla** (free and most popular FTP client)
- **WinSCP, Cyberduck**
- **Steps to Deploy a Website using FTP:**
- **Install FileZilla** on your computer.
- **Get FTP credentials** from your hosting provider:
 - Host (e.g., ftp.yourdomain.com)
 - Username
 - Password
 - Port (usually 21)
- **Connect to the server** using FileZilla.
- **Upload your project files** (e.g., index.php, style.css) to the **public_html** folder.
- **Access your website** in a browser using your domain name.
- Your website is now **live on the Internet!**

PHP Framework Using WordPress

- **What is WordPress?**
- **WordPress is a free and open-source Content Management System (CMS).**
- **It allows you to create, manage, and publish websites easily — without coding.**
- **It is built using PHP and MySQL.**

Why Use WordPress?

Feature

Easy to Use

Explanation

No need to know coding — everything is GUI-based.

Customizable Themes

Choose from thousands of free and paid themes to change website design.

Plugins

Add extra features (e.g., contact forms, sliders, SEO tools) without programming.

Dynamic Website

Supports blogs, portfolios, e-commerce, and full websites.

Secure and Updatable

Regular updates and plugins keep it secure.

How WordPress Works

- WordPress uses:
- **PHP** → To process and display content dynamically.
- **MySQL** → To store your website data (posts, users, settings).
- **Apache/Nginx Server** → To serve your website on the web.

WordPress Architecture (Simple View)

- User → Browser → WordPress (PHP Files)
- ↓
- MySQL Database

- When a user opens your site:
- WordPress PHP code runs on the server.
- It fetches data (posts, pages, etc.) from the **MySQL database**.
- The data is displayed to the user through a theme.

Installing WordPress (Locally using XAMPP)

- **Step 1: Install XAMPP**

- XAMPP includes **Apache, PHP, and MySQL**.
- Start **Apache** and **MySQL** from the XAMPP Control Panel.

- **Step 2: Download WordPress**

- From <https://wordpress.org/download/>
- Extract it into the htdocs folder (e.g., C:\xampp\htdocs\wordpress).

- **Step 3: Create Database**

- Go to **phpMyAdmin** → Create a new database (e.g., wordpress_db).

- **Step 4: Run WordPress Setup**

- Visit <http://localhost/wordpress> in your browser.

- Fill in details:

- Database Name: wordpress_db
- Username: root
- Password: *(leave empty for XAMPP)*
- Site Title, Username, Password, Email.

- **Step 5: Log in**

- Visit <http://localhost/wordpress/wp-admin>

- Enter the admin credentials and start customizing your website!

WordPress Key Components

Component

Dashboard

Posts

Pages

Themes

Plugins

Widgets

Menus

Description

The admin panel to manage your site.

Used for blog entries or updates.

Used for static content (e.g., About Us, Contact).

Control how your website looks.

Add extra features like forms, galleries, etc.

Add small features (e.g., search bar, calendar).

Manage navigation links.

Deploying WordPress Website to Web

- **Export Database** from local phpMyAdmin.
- **Upload WordPress files** to your hosting using FTP (FileZilla).
- **Import Database** into hosting phpMyAdmin.
- **Update wp-config.php** with new database details.
- Visit your domain — your site is live!

- **Steps to Create a Website using WordPress:**
- **Download & Install WordPress** from wordpress.org.
- **Set up the database** using phpMyAdmin.
- **Run installation wizard** (localhost/wordpress).
- **Login** to admin panel: localhost/wordpress/wp-admin.
- **Choose a theme, add pages, and publish your site.**

- **Advantages:**

- Easy to use and beginner-friendly
- SEO and mobile friendly
- Thousands of free themes and plugins
- Used for blogs, portfolios, and business sites

Concept	Description	Example / Tool
Domain	Website name / address	www.example.com
Hosting	Stores website files	Hostinger, GoDaddy
Deployment (FTP)	Uploads files to web server	FileZilla
WordPress (PHP Framework)	CMS built on PHP + MySQL	wordpress.org